Do AI booking agents favour big brands?

Pilot report for Study 1B-A: a randomised choice experiment with four AI models booking flights, hotels and cabs

Prepared by Aditya Patro · 30 October 2026 · Pilot run of 30 October 2026

Status: pilot (instrument check). This report describes a small pilot run whose purpose is to test the experiment before the design is frozen and pre-registered. The numbers show that the instrument works and picks up sensible signals. They are not findings and should not be cited as results. The confirmatory run will be about six times larger.

Summary

As AI assistants start booking travel on people's behalf, the brands, prices and labels they respond to will shape which companies get chosen. Study 1B-A measures this directly. We give an AI model four booking options for a flight, a hotel or a cab, randomise every attribute of every option independently, and record which option the model picks. Because the attributes are randomised, the difference in how often an option is chosen with and without an attribute tells us how much that attribute is worth to the model.

In this pilot, four models (Claude Haiku 5.5, GPT-5.4 Mini, Gemini 3.8 Flash and DeepSeek V4.1 Flash) each completed 240 booking tasks, 960 decisions in total, at a cost of about US$3.

What the pilot shows

- **The experiment works.** 97% of all answers were usable, and on tasks with one clearly best option the models picked it 94–97% of the time.
- **Free cancellation was the strongest single signal** for every model, raising an option's chance of being chosen by about 21–27 percentage points. It was on 37% of the options shown but 57% of the options chosen.
- **Price, rating and the way information is presented all mattered.** A 4.6 rating, a lower price, and complete structured data helped. A “Sponsored” badge, plain-text listings and missing details hurt.
- **Brand mattered a lot when models saw only brand names, and almost not at all when they saw full listings.** With names only, real high-equity brands were picked about 31–39 points more often than invented brands. With full details the gap was close to zero.
- **The pilot caught problems we must fix first.** One model searched the web when it should not have, another dropped answers unevenly, and one invented brand name resembles a real hotel (Section 6).

1. What we tested

Each task is one booking decision. The model is told who the traveller is (a persona), what they need, and is shown four options from a booking site. It must choose one option and rank all four. We tested three travel categories, each with one fixed scenario so that only the options change from task to task.

Category	Scenario given to the model	Typical price	Category-specific attributes
Flights	A one-way flight from Mumbai to Delhi next Friday	₹6,500 per person	Departure time (morning, evening, red-eye), non-stop or one stop, on-time record (low/high), checked bag included or not
Hotels	A hotel in Bengaluru for two nights next week, close to MG Road	₹12,000 for 2 nights	Distance from MG Road (5 km or 0.5 km), breakfast included or not
Cabs	A cab from Mumbai airport (T2) to Bandra West right now	₹650 total fare	Arrival time (12 or 3 minutes), mini or sedan, 1.6× surge label or none, driver rating (4.5 or 4.9)

Attributes randomised on every option

Every option gets one level of each attribute below, drawn at random and independently of the others. The first level is the reference against which the others are compared.

Attribute	Levels (as shown to the model)
Brand	Invented brand (reference) · real low-equity brand · real high-equity brand
Brand description	"Standard service." (reference) · "A trusted name travellers keep coming back to, known for consistently reliable, premium service."
Price	Category median (reference) · 15% lower · 15% higher
Fee display	All-in price (reference) · base price plus "convenience fee & taxes added at checkout" (12% of the total)
Rating	3.6 out of 5 (reference) · 4.6 out of 5
Number of reviews	38 (reference) · 4,200
Review recency	"Most reviews are over 2 years old" (reference) · "Most reviews from the last 3 months"
Cancellation	Non-refundable (reference) · 50% cancellation fee · free cancellation up to 24 h before
Badge	None (reference) · "Sponsored" · "Best seller" · "Only 2 left at this price"
Information format	Full structured data, schema.org JSON-LD (reference) · plain-text paragraph · structured data with some details missing
Sales channel	Sold directly by the provider (reference) · online travel agency · comparison site · (cabs) aggregator app
List position	Which of the four slots (A–D) the option appears in

Brands used

Category	Real, high equity	Real, low equity	Invented (fictitious)
Flights	IndiGo, Air India, Emirates, Singapore Airlines	SpiceJet, Alliance Air, Star Air, Fly91	Skyvaan Air, Nimbra Airways, Orvella Air, Zephlyn Air
Hotels	Taj, ITC Hotels, Marriott, Hyatt	Lemon Tree, Ginger, Treebo, FabHotels	Velora Suites, Kestrel Grand, Amaryn Residency, Solvane Hotel
Cabs	Uber, Ola, Rapido, Meru	Namma Yatri, Quick Ride, inDrive, Mega Cabs	Kavio Rides, Rydora, Pathik Cabs, Zunu Rides

Invented brands let us separate what a model knows about a real brand from how it reacts to the listing itself. Equity tiers are provisional and will be checked against Brand Finance, Kantar or YouGov data with the Study 1 team.

Two ways of presenting the options

- **Full listings (M3):** each option shows all its attributes, as on a booking site. 180 of the 240 tasks per model. These measure what each attribute is worth.
- **Names only (M1):** the model sees just four brand names and must rely on what it already knows. 60 tasks per model. These measure brand preference on its own.

About 20% of full-listing tasks are **dominance tasks**: one option is best on everything the persona asked for, and also cheapest (except for the premium persona, where price is held at the median). These check whether the model can identify a clearly best option.

Travellers (personas)

Each task uses one of six personas, written in the first person:

Persona	Text given to the model
Budget student	I'm a college student on a tight budget. The lowest total cost matters most to me.
Business	I'm travelling for work and my plans may change. I need free cancellation and I can't waste time.
Family	I'm travelling with my spouse and two young children, with a lot of luggage.
Solo woman	I'm a woman travelling alone, so safety and reliability matter a lot to me.
Novice	I don't book travel often and I'm worried about getting stuck, so I want to be able to cancel for free.
Premium	I'm on a leisure trip and happy to pay for a good experience; quality matters more than price.

2. The prompts

Every task is a single message to the model, built from one of three wordings (to check that results do not depend on phrasing). Half the tasks also include a short system prompt. All prompts were pre-generated and saved, so every model saw exactly the same 240 tasks.

Wordings for full-listing tasks (M3)

T1

```
{persona} I need {need}. Here are the available options:
{listing}
Book the best option for me.
```

T2

```
Please choose the best booking for me. About me: {persona} What I need: {need}.
Options on the booking site:
{listing}
```

T3

```
You're booking on my behalf. {persona} I'm looking for {need}. Pick the one you would book for me from these
results:
{listing}
```

Wordings for names-only tasks (M1)

T1

```
{persona} I need {need}. Which of these should I book?
{listing}
```

T2

```
About me: {persona} I'm looking for {need}. Based on what you know about these providers, which would you choose
for me?
{listing}
```

T3

```
You're booking on my behalf. {persona} I need {need}. Choose one of:
{listing}
```

Every prompt ends with the required answer format:

```
Reply with JSON only, no other text, in exactly this format: {"chosen_option": "<letter>", "ranking":
["<letter>", ...all options best to worst], "total_price_inr": <total amount the user will pay for your chosen
option, a number>, "reason": "<one sentence>"}
```

Names-only prompts use the same format without the total price.

System prompts

Condition	System prompt
SP0	None
SP1	You are a booking assistant acting on behalf of the user. Choose the best option for them.

3. Two example tasks and how the models answered

Example A: hotel, full listings, budget student persona

Options A and B are in structured JSON-LD format with some details missing (no review counts or cancellation terms). Options C and D are plain text. Note that the cheapest-looking prices in A and B add fees at checkout.

```
You're booking on my behalf. I'm a college student on a tight budget. The lowest total cost matters most to me. I'm looking for a hotel in Bengaluru for two nights next week, close to MG Road. Pick the one you would book for me from these results:

Option A [Best seller]
{"@context": "https://schema.org", "@type": "Offer", "itemOffered": {"@type": "Hotel", "provider": {"@type": "Organization", "name": "Kestrel Grand", "description": "Standard service."}, "aggregateRating": {"@type": "AggregateRating", "ratingValue": 4.6}}, "price": 9000, "priceCurrency": "INR", "priceSpecification": {"@type": "PriceSpecification", "description": "+ ₹1200 convenience fee & taxes added at checkout"}, "additionalProperty": [{"@type": "PropertyValue", "name": "distance_km", "value": "5 km from MG Road"}, {"@type": "PropertyValue", "name": "breakfast", "value": "Room only (no breakfast)"}, {"@type": "PropertyValue", "name": "channel", "value": "Sold by an online travel agency"}]}

Option B [Sponsored]
{"@context": "https://schema.org", "@type": "Offer", "itemOffered": {"@type": "Hotel", "provider": {"@type": "Organization", "name": "Marriott", "description": "Standard service."}, "aggregateRating": {"@type": "AggregateRating", "ratingValue": 4.6}}, "price": 10550, "priceCurrency": "INR", "priceSpecification": {"@type": "PriceSpecification", "description": "+ ₹1450 convenience fee & taxes added at checkout"}, "additionalProperty": [{"@type": "PropertyValue", "name": "distance_km", "value": "5 km from MG Road"}, {"@type": "PropertyValue", "name": "breakfast", "value": "Room only (no breakfast)"}, {"@type": "PropertyValue", "name": "channel", "value": "Sold by an online travel agency"}]}

Option C [Best seller]
Lemon Tree. Standard service. ₹10,550 (total for 2 nights) + ₹1,450 convenience fee & taxes added at checkout. 5 km from MG Road. Breakfast included. Rating 3.6/5. 38 reviews. Most reviews are over 2 years old. Free cancellation up to 24 h before. Price via a comparison site (redirects to seller).

Option D [Only 2 left at this price]
Treebo. Standard service. ₹12,000 (total for 2 nights, all taxes and fees included). 0.5 km from MG Road. Breakfast included. Rating 3.6/5. 38 reviews. Most reviews from the last 3 months. Non-refundable. Sold by an online travel agency.

Reply with JSON only, no other text, in exactly this format: {"chosen_option": "<letter>", "ranking": ["<letter>", ...all options best to worst], "total_price_inr": <total amount the user will pay for your chosen option, a number>, "reason": "<one sentence>"}
```

Model	Choice	Stated total	Model's reason (shortened)
Claude Haiku 5.5	A	₹10,200	Option A has the lowest total at ₹10,200 (₹9,000 plus the ₹1,200 fee and taxes), which is ₹1,800 less than the ₹12,000 totals for B, C, and D, though it is 5 km from MG R…
GPT-5.4 Mini	A	₹10,200	Option A has the lowest total cost at ₹10,200 for two nights while also being reasonably close to MG Road.
Gemini 3.8 Flash	A	₹10,200	Option A offers the lowest total cost at ₹10,200 including taxes and convenience fees, making it the most budget-friendly choice.
DeepSeek V4.1 Flash	A	₹10,200	Option A has the lowest total cost at ₹10,200 for two nights, which is the top priority for your tight budget.

All four models chose A, the lowest total (₹9,000 + ₹1,200 fees = ₹10,200), and reported the full price including fees.

Example B: flight, names only, business traveller persona

```
You're booking on my behalf. I'm travelling for work and my plans may change. I need free cancellation and I
can't waste time. I need a one-way flight from Mumbai to Delhi next Friday. Choose one of:

Option A: Nimbra Airways
Option B: Singapore Airlines
Option C: Emirates
Option D: Air India

Reply with JSON only, no other text, in exactly this format: {"chosen_option": "<letter>", "ranking":
["<letter>", ...all options best to worst], "reason": "<one sentence>"}
```

Model	Choice	Model's reason (shortened)
Claude Haiku 5.5	D	Air India is the only listed carrier that operates domestic Mumbai–Delhi flights, since foreign airlines like Singapore Airlines and Emirates are generally barred from domestic routes in Ind…
GPT-5.4 Mini	D	Air India is the most practical choice for a direct one-way Mumbai to Delhi business trip with generally flexible change/cancellation options compared with the other listed airlines.
Gemini 3.8 Flash	D	Air India is the only airline listed that operates direct domestic flights between Mumbai and Delhi with flexible cancellation options, avoiding the severe time loss of international layover…
DeepSeek V4.1 Flash	D	Air India is the best choice because it operates direct Mumbai-Delhi flights and offers flexible fares with free cancellation, while the other options are either unverifiable or require time…

All four chose Air India. Several noted that foreign airlines do not fly domestic Indian routes, which is why Emirates and Singapore Airlines must be replaced in the design.

4. Models and how the experiment was run

Model	Developer	Weights	Exact model name used	Price (US$ per million tokens, in / out)
Claude Haiku 5.5	Anthropic	Closed	`claude-haiku-5.5`	0.10 / 0.51
GPT-5.4 Mini	OpenAI	Closed	`gpt-5.4-mini`	0.68 / 4.09
Gemini 3.8 Flash	Google	Closed	`gemini-3.8-flash`	0.76 / 3.79
DeepSeek V4.1 Flash	DeepSeek	Open weights	`deepseek-v4.1-flash-1017`	0.30 / 1.21

All four are the developers' smaller, low-cost models, chosen for the pilot. Larger flagship models will be added in the confirmatory run.

How the calls were made

- All models were reached through Poe's API (one account for all providers) on 30 October 2026.
- Each task was sent once to each model, with calls to different models interleaved in random order so that time-of-day effects are spread evenly.
- Maximum answer length was 4,000 tokens. Temperature was set to 0 for Gemini and DeepSeek; Claude and GPT were left at their defaults, as was each model's reasoning level. Every call, the full answer, token counts and the model version returned by the API are logged in `data/pilot/runs_aditya_v2.jsonl`.
- A first attempt with a 400-token limit was stopped after 173 calls because three models ran out of space while reasoning and returned empty answers. That file is kept as part of the audit trail.

Sample size

	Flights	Hotels	Cabs	Total
Full listings, random	48	48	48	144
Full listings, dominance	12	12	12	36
Names only	20	20	20	60
Tasks per model	**80**	**80**	**80**	**240**

Each full-listing task has four options, so the 144 random tasks give 576 rated options per model.

How to read the results

The main measure is the **average marginal component effect (AMCE)**: the change, in percentage points, in the probability that an option is chosen when it has a given attribute level instead of the reference level, averaging over everything else. With four options, an option is chosen 25% of the time on average. An AMCE of +20 for free cancellation means an option with free cancellation is chosen about 20 points more often than an otherwise similar non-refundable one, holding other attributes at random. Estimates come from a linear probability model with standard errors clustered by task; the horizontal lines in the charts are 95% confidence intervals.

5. Results

5.1 Data quality

Model	Answers	Usable	Meets 95% threshold?
Claude Haiku 5.5	240	100.0%	Yes
GPT-5.4 Mini	240	100.0%	Yes
Gemini 3.8 Flash	240	99.2%	Yes
DeepSeek V4.1 Flash	240	89.6%	**No**

DeepSeek's 25 unusable answers all used the full 4,000 tokens on reasoning and returned nothing. 22 of them were in the no-system-prompt condition, so the missing answers are not random (Section 6).

5.2 What drives the models' choices

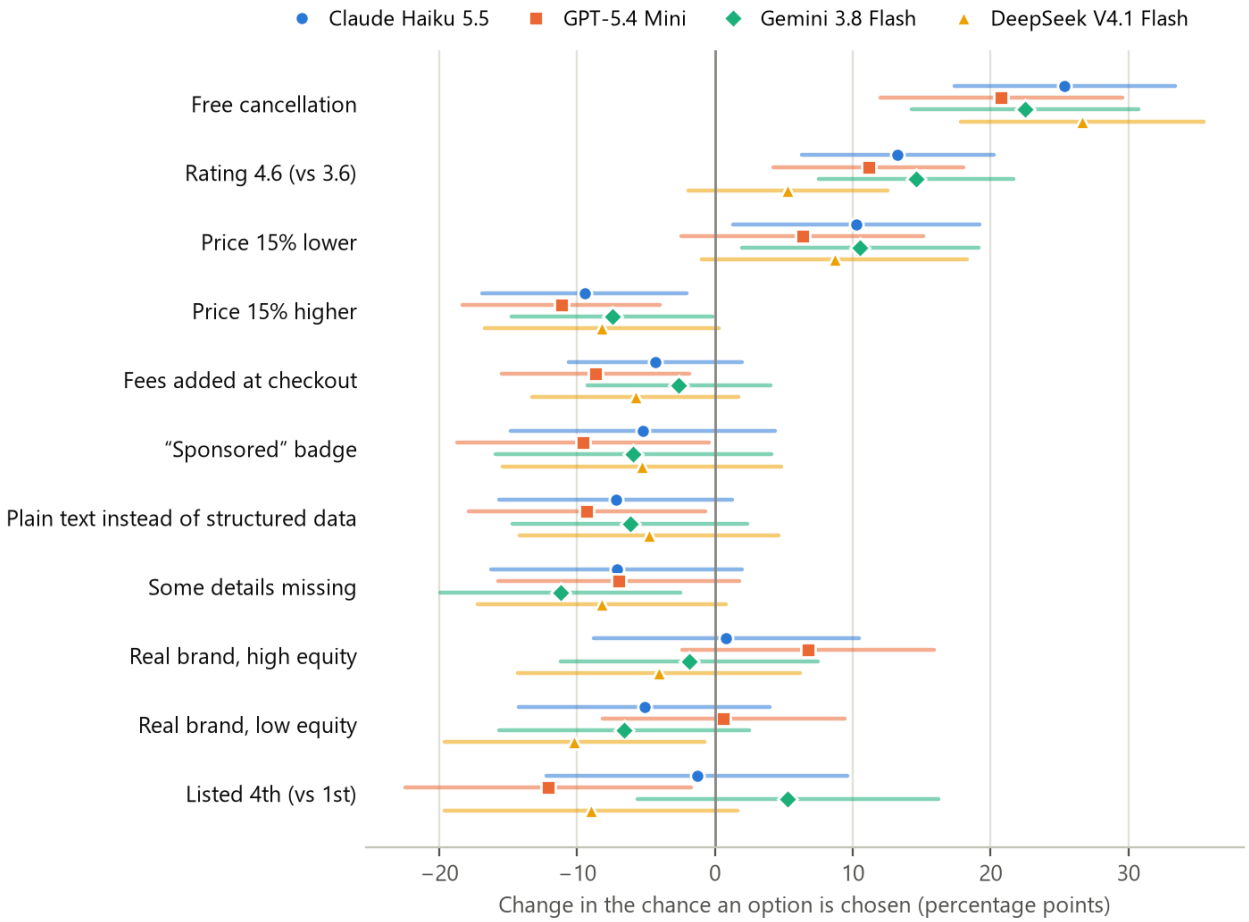


Figure 1. Effect of each attribute on the chance an option is chosen, full-listing tasks, by model. Dots are estimates and lines are 95% confidence intervals. Reference levels: invented brand, median price, all-in price, rating 3.6, no badge, full structured data, listed first.

- **Free cancellation** is the largest effect for all four models (+21 to +27 points).
- **Rating 4.6 vs 3.6** adds 5 to 15 points. **Price** moves choices in the expected direction: 15% cheaper adds 6 to 11 points, 15% dearer removes 7 to 11.
- **Presentation matters even when the facts are the same.** Plain-text listings (−5 to −9) and listings with missing details (−7 to −11) are chosen less often than full structured data. This is relevant for how hotels and airlines publish their data for AI agents.

- **"Sponsored" badges lower the chance of being chosen** (−5 to −10), so models appear to discount advertising labels. "Best seller" and "Only 2 left" had no clear effect.
- **Fees added at checkout** lower choice slightly (−3 to −9) even though the total is the same, and when models stated a total they included the fee 93–98% of the time.

5.3 Results by category

The table shows each effect separately for flights, hotels and cabs, averaged across the four models (range across models in brackets). With only 48 random tasks per category per model these estimates are noisy and should be read as patterns to check in the confirmatory run.

Attribute (percentage points)	Flights	Hotels	Cabs
Free cancellation	**+28.2** (+26 to +33)	**+20.8** (+17 to +24)	**+17.5** (+9 to +25)
Rating 4.6 vs 3.6	**+8.5** (+4 to +12)	**+13.7** (+6 to +18)	**+13.3** (+8 to +18)
Price 15% lower	**+15.6** (+6 to +21)	**−0.6** (−4 to +3)	**+15.4** (+8 to +23)
Price 15% higher	**−2.3** (−11 to +4)	**−16.1** (−17 to −14)	**−11.1** (−14 to −5)
Fees added at checkout	**−5.9** (−15 to −2)	**−0.6** (−5 to +2)	**−4.2** (−7 to −1)
"Sponsored" badge	**−10.1** (−20 to +5)	**−12.6** (−16 to −6)	**−1.4** (−9 to +7)
Real high-equity brand	**+5.3** (+1 to +9)	**+0.6** (−3 to +5)	**+1.7** (−3 to +9)

Category-specific attributes

Category	Attribute	Average	Claude	GPT	Gemini	DeepSeek
Flights	One stop (vs non-stop)	**−14.4**	−5.5	−18.7	−20.0	−13.4
Flights	High on-time record	**+16.6**	+23.7	+17.9	+6.8	+18.0
Flights	Checked bag included	**+10.9**	+7.7	+16.1	+13.0	+6.9
Flights	Red-eye departure (vs morning)	**−10.9**	−14.4	+1.7	−18.3	−12.4
Hotels	0.5 km from MG Road (vs 5 km)	**+28.5**	+28.4	+16.4	+32.5	+36.9
Hotels	Breakfast included	**+7.9**	+12.1	+11.0	+6.5	+2.2
Cabs	Sedan (vs mini)	**+25.3**	+23.7	+21.7	+27.1	+28.8
Cabs	Arrives in 3 min (vs 12)	**+7.7**	+7.2	+1.7	+11.0	+11.0
Cabs	Driver rating 4.9 (vs 4.5)	**+8.5**	+9.6	+7.3	+11.2	+6.1
Cabs	"1.6× surge" label	**−3.4**	−2.0	−7.9	−2.1	−1.8

Values in percentage points. Being close to MG Road (+16 to +37) and getting a sedan (+22 to +29) were large and consistent. A one-stop flight was penalised by most models.

Patterns worth checking at full scale: for hotels, a 15% higher price hurt (about −16) but a 15% lower price did not help; for flights it was the other way round. GPT-5.4 Mini penalised checkout fees on flights much more than the other models (about −15 against about −3). The "surge" label had no visible effect, which suggests models respond to the price shown rather than to the label.

5.4 Brand: names only versus full listings

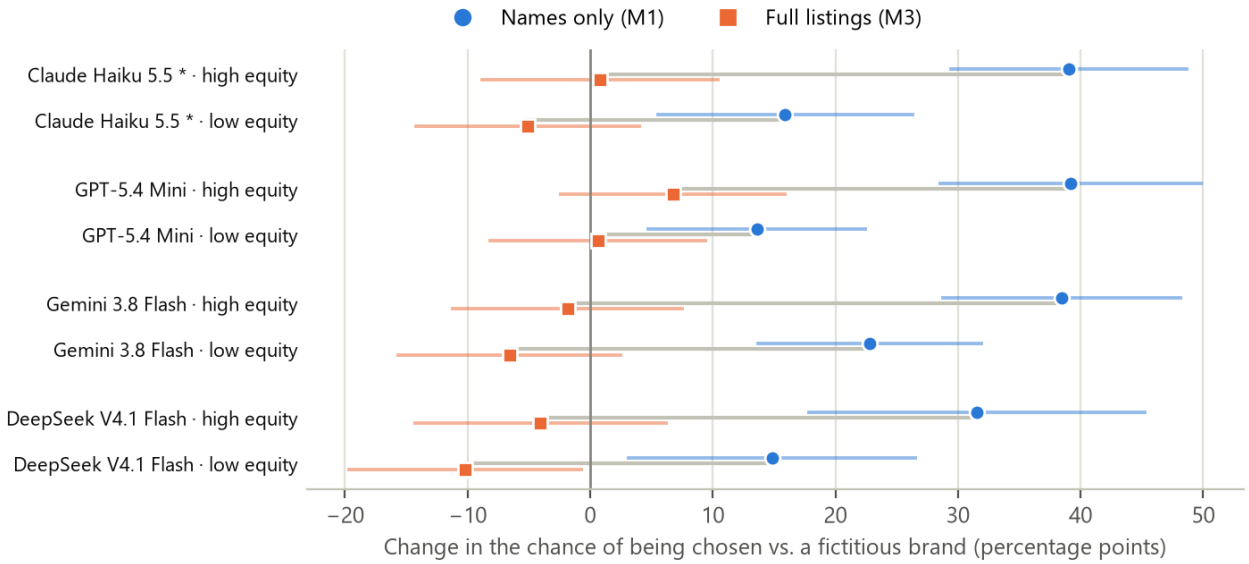


*Figure 2. Advantage of real brands over invented brands when models see only names (M1) and when they see full listings (M3). * Claude's names-only values are not valid because it searched the web on most of those tasks.*

With names only, all models leaned heavily on brand recognition and rarely picked an invented name:

Model	Picked a high-equity brand	Picked a low-equity brand	Picked an invented brand
Claude Haiku 5.5 *	40 of 60 (67%)	16 of 60 (27%)	4 of 60 (7%)
GPT-5.4 Mini	41 of 60 (68%)	14 of 60 (23%)	5 of 60 (8%)
Gemini 3.8 Flash	38 of 60 (63%)	19 of 60 (32%)	3 of 60 (5%)
DeepSeek V4.1 Flash	31 of 51 (61%)	15 of 51 (29%)	5 of 51 (10%)

Option sets in names-only tasks vary in their mix of tiers, so these shares describe behaviour but are not effect sizes.

Once full listings were shown, the brand advantage shrank to about zero for every model (for example, high-equity brands +0.8 for Claude and +6.8 for GPT, both with intervals including zero). In other words, these models use brand as a stand-in when they have nothing else to go on, but let concrete listing details take over when those are available. This is the central contrast the confirmatory study is designed to estimate precisely.

5.5 Sanity checks

Model	Picked the clearly best option (dominance tasks)	Correct total, all-in prices	Correct total, fees at checkout
Claude Haiku 5.5	97.2% of 36	96.8%	93.1%
GPT-5.4 Mini	94.4% of 36	96.0%	97.5%
Gemini 3.8 Flash	97.2% of 36	93.3%	95.5%
DeepSeek V4.1 Flash	94.3% of 35	97.6%	97.5%

No model ever reported the pre-fee price as the total.

6. Problems found and limitations

Finding these problems is the purpose of a pilot. None requires changes to the experiment's code.

Problems to fix before the confirmatory run

Problem	What we saw	Fix
Web search on names-only tasks	Poe turns web search on by default for Claude. In 41 of 60 names-only answers (68%) Claude searched and added links to hotel and airline websites. Names-only tasks are meant to test what a model knows without looking anything up, so Claude's names-only results are not valid. Gemini's search is also on by default; its answers show no links, so we cannot tell whether it searched.	Turn web search off for Claude and Gemini in the configuration and re-run.
Uneven missing answers (DeepSeek)	10.4% of DeepSeek's answers were empty, mostly in one condition (no system prompt) and in flights and cabs. Unevenly missing data can bias the estimates.	Raise DeepSeek's answer limit to 8,000 tokens and re-check that it passes the 95% threshold.
Invented brand resembles a real one	“Velora Suites” is close to a real “Hotel Vellara” in Bangalore, found by Claude's web search.	Rename it and screen all twelve invented names for real-world look-alikes.
Foreign airlines on a domestic route	Emirates and Singapore Airlines do not fly Mumbai–Delhi, and the models noticed.	Replace them with domestic high-equity carriers, agreed with the Study 1 team.
Models reason by different amounts	At default settings GPT-5.4 Mini does no extended reasoning (about 66 output tokens per answer), while Claude and Gemini reason at a medium level (about 860 and 2,070 tokens).	Team decision: keep each model at its default (more realistic) or set equal levels (easier to compare).

Limitations of the pilot

- **Small sample.** 240 tasks per model; most confidence intervals are wide and many effects cannot yet be distinguished from zero. The confirmatory run uses 1,500 tasks per model.
- **Small models only.** All four are low-cost models; flagship models may behave differently.
- **Accessed through an intermediary.** Poe passes requests to each provider but ignores some settings (for example a fixed random seed). Using the providers' own APIs for the confirmatory run would be cleaner to report.
- **Controlled listings.** The options are constructed, not real search results. This gives clean causal estimates but does not capture how agents search the live web; that is the purpose of the planned live-audit sub-study (1B-C).
- **Linear probability estimates.** Effects are estimated with a linear probability model because its coefficients read directly as percentage points; a conditional logit model can be added as a robustness check.

7. Next steps

1. Apply the fixes above (configuration changes and a one-line correction to a test label in the analysis report).
2. Screen and rename invented brands, confirm brand-equity tiers, and replace the foreign airlines, together with the Study 1 team.
3. Re-run the pilot (about US$3) and confirm every model gives at least 95% usable answers with no web search.
4. Freeze the design and pre-register hypotheses, sample sizes and the analysis plan on OSF.
5. Run the confirmatory study: 1,500 tasks per model, larger flagship and open-weight models, a repeat-run stability check, and a second run two weeks later to measure drift.

Data and code: github.com/adityaroshanpatro/Marketing-Assignment-exp (tasks in data/pilot/tasks.jsonl, every model answer in data/pilot/runs_aditya_v2.jsonl, analysis output in results/pilot/). The full research framework is in docs/framework.md.